

ENEL372S2 Homework 2

1. I am designing a caving light, which uses a buck converter to drive some high power LEDs. The supply voltage can range from 4.8 volts (NiMH batteries) to 7.2 volts (Li-ion batteries), and the LED voltage is 3.6 volts at 900mA. The inductor peak current is 1 Amp, and the switching frequency is 250kHz. Figure 1 shows the circuit.

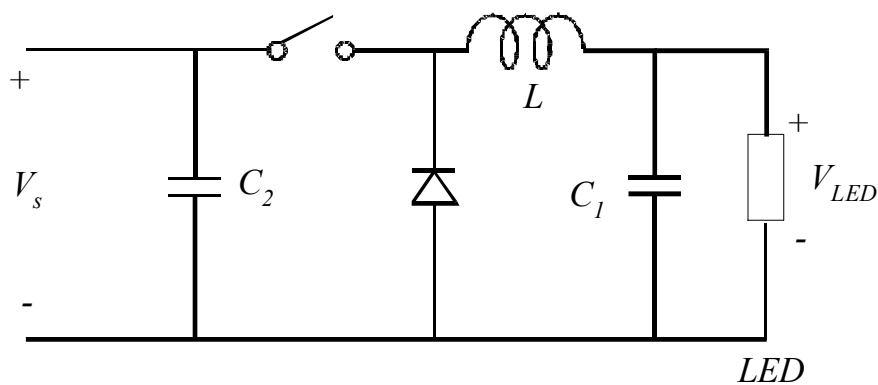


Figure 1. Buck converter circuit

- (i) Sketch waveforms showing inductor voltage, inductor current, capacitor C_1 current and capacitor C_1 voltage ripple. Clearly show their alignment in time.
- (ii) Derive an expression for the minimum inductance required to limit the peak inductor current to 1 Amp, in terms of the specifications given above. Calculate the inductance value.
- (iii) The supply batteries are most efficient when supplying a constant average current. Assuming a steady current flow from the supply, calculate the capacitance value (C_2) to limit the local supply voltage ripple to 100mV peak to peak.